

FEATURES

Speeds

Until last year, we thought we'd never see a DVD-Writer that would be able to breach the writing speed of 16x for DVDs. But this time round, we received four drives capable of doing just that! Even though the DVD Forum (originally the DVD Consortium) still specifies 16x as the maximum speed of a DVD, drives from LG, Samsung, and Sony sported 18x write speeds for DVD-R and DVD+R. The Lite-On Super AllWrite LH-20A1P surprised us the most when we found that it could manage 20x!

The DVD rewriting speeds put out by the drives were similar: 8x for DVD+RW and 6x for DVD-RW. All the drives are capable of writing to dual-layer media, and most could do so at 8x to DVD+R DL—with the exception of the LG, capable of 10x. Similarly, with the exception of the Super AllWrite LH-16A1P and the Lite-On LightScribe SHM-165H6S, which could write to DVD-R DL at only 4x, the rest were twice as fast at 8x. For those heavily into DVD movies, these high-speed drives are just right.

DVD-RAM is a media that presents itself as a hard drive. You can copy data to it, delete it, overwrite it, rename it, or do any other file operation just as if you would do with files on your hard drive. The good thing is that you don't need DVD-Writing software to achieve this. This is especially useful for video editing professionals and the like.

During last year's DVD-Writer test, we received only a couple of drives capable of writing to DVD-RAM, but this time the picture was reversed: only one of the drives, the Asus DRW-1608P2, was *not* capable of writing to DVD-RAM.



Lite-On Super AllWrite LH-20A1P
Speed *can* come cheap!

However, even this drive was capable of reading from DVD-RAM, albeit at a crawling 2x. While the Super AllWrite LH-16A1P and LightScribe SHM-165H6S from Lite-On could write to DVD-RAM at 5x, the rest were capable of 12x.

48x was the speed at which most of the drives could write to CD-R, while the ageing Asus 1608P2 could only manage 40x. Though the Lite-On Super AllWrite LH-16A1P and LightScribe SHM-165H6S, which could write at 24x to CD-RW, the Sony AW-G170A whizzed at 40x. The rest could do 32x.

Other Features

Some of the drives had noteworthy features:

LightScribe allows you to burn a CD or DVD label using the same drive that you use to write data. You can create your own label using images or text or both, and burn it to the face of the media. The drive employs an additional laser head for this. You require LightScribe media, though. We received two drives capable of LightScribe: the Asus DRW-1612BL and the Lite-On LightScribe SHM-165H6S. Because LightScribe media is scarce, this feature was not awarded too many points, but still, the feature shouldn't hurt!

Setting "Book Type" of a DVD means you can make the DVD you create appear like a stamped DVD-ROM to any standalone DVD player. Some DVD players have an issue playing DVD video discs created using DVD-Writers, and this feature aims to address that issue. The Asus DRW-1612BL, LG GSA-H12N, Lite-On SHM-165H6S, and the Samsung WriteMaster SH-182D had the feature of setting the media identification bit in the recordable DVD to make it appear as a stamped DVD-ROM.

The drives we received this time were all ruggedly built, unlike during the last comparison, and we must applaud the thought that the manufacturers have put into ensuring the longevity of

How We Tested

The DVD-Writers and Combo drives were tested on a machine that comprised an AMD Athlon64 3800+ processor running at a stock speed of 2.0 GHz, a WinFast 6150K8MA motherboard, 1 GB of 400 MHz Corsair XMS DDR RAM, and a Seagate Barracuda 120 GB SATA150 hard drive. The test machine was powered by an Antec Neo480 480-watt power supply. Windows XP Professional SP2 with the latest drivers was installed afresh on the test machine. Nero 7 Premium Edition was installed for CD and DVD burning. The drives were set as master and connected to the primary IDE channel. The test process consisted of two parts: writing to and reading from different types of media.

The Reading Tests

Nero CD-DVD Speed was used to note the data transfer rates, access/seek times, CPU utilisation, burst rate, etc. for CDs as well as DVDs. This comes bundled with Nero 7 Premium.

SiSoft Sandra Pro Business 2007, which is a system-wide performance benchmark as well as diagnostic utility, was used to measure the DVD read speeds and access times.

dbPowerAmp, an audio format conversion and audio CD ripping tool, was used to rip three selected audio tracks from our test audio CD. The tracks were ripped to 44 kHz MP3 format at 128 Kbps, and the time was logged.

DVD Decrypter is a tool used to rip DVD video files to a hard drive. This was used to rip a selected chapter (a .VOB file, 1 GB) from our test DVD to the hard drive. The time taken was noted.

The Writing Tests

These included logging the time taken to write to CD-R, CD-RW, DVD-R, DVD+R, DVD-RW and DVD+RW media. 700 MB of assorted data—the latest *Digit* CD—was used to write to CD-R, CD-RW and DVD-RW, and the time was logged. Sequential data consisting of a 700 MB file—the same *Digit* CD zipped without compression—was used to write to a CD-R. Similarly, assorted and sequential data of 4.7 GB was used to write to DVD-R and DVD+R media, and the time was noted.

Combo drives went through all the above except for the DVD tests.

The Media

The media used were 52x CD-R from Moser Baer, 10x CD-RW from HP, 16x DVD-R from Sony, 16x DVD+R from Sony, and 4x DVD-RW from HP. A point to note is that some of the DVD-Writers can burn DVDs at speeds in excess of 16x, but since such media is unavailable, the 16x media was burnt at the highest supported speed.

Other Features

Features such as support for various DVD types, LightScribe, capability to set Book Type, tray build quality, silent operation, and bundled accessories such as a data cable, manual, media, bezels, etc. were also noted and rated to derive the final score. Note that the specified read and write speeds were not considered; we only rated the drives based on real world read and write speeds.